
MR DICOM Visualization and Catheter Guidance

Can mixed reality give surgeons the spatial awareness they need to perform remote stroke surgery?

Key solution: Real-time volumetric MR rendering of patient CT data with graph-constrained catheter simulation on Meta Quest 3

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Updates

- **Rebuilt navigation from scratch**

- graph-based, constrained to real vessel centerlines

- **Full two-phase simulation**

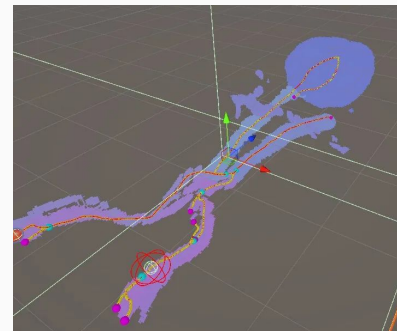
guidewire → catheter → stent
deploy → extraction

- **Haptics**

8-event haptic system throughout the procedure

- **Fork puzzle**

twist guidewire to unlock branches at bifurcations



- **Fluoroscopy**

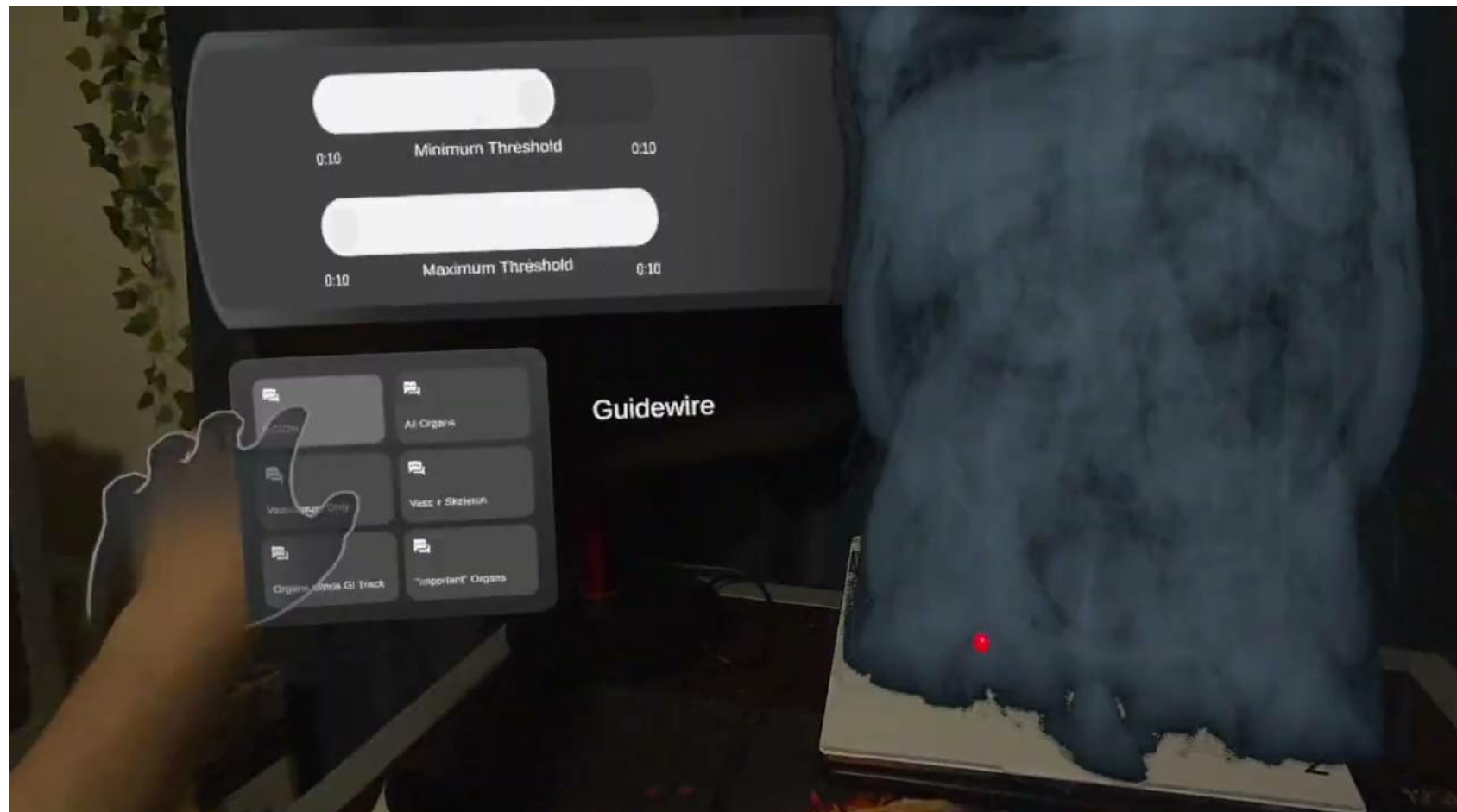
hold A for X-ray view, clot visible as filling defect only in fluoro



- **State Flow**

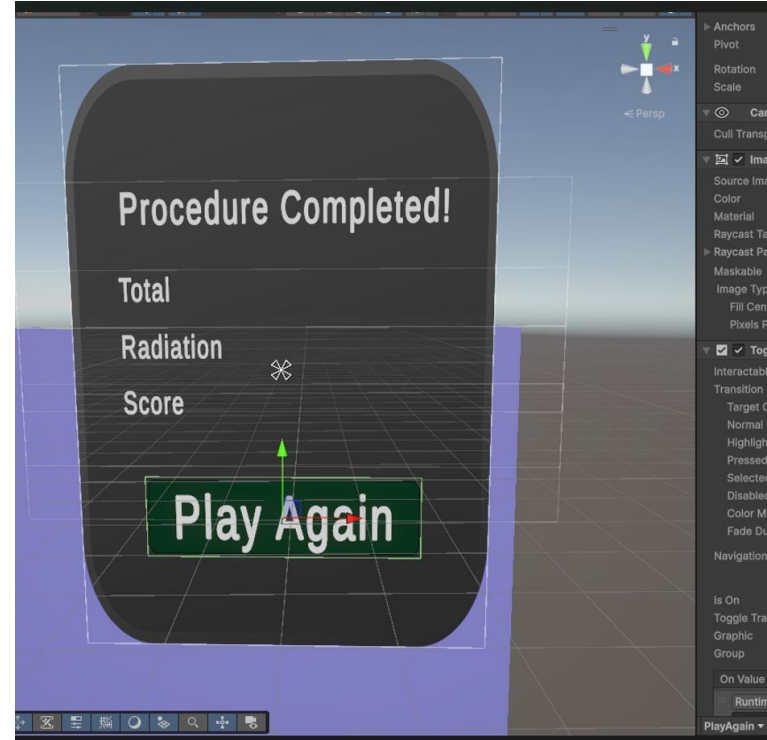
Complete state machine driving all phase transitions

Demo



Work in progress

- Haptics needs adjustments
- Results UI should be fixed and connected
- Need to fix the bug with the fork alignment arrows
- UI for instructions and progress update



Reflection

- **Biggest mistake:** started with physics-based free steering, spent significant time debugging collisions and drift before realizing graph-based constraint was the right foundation
- Early raymarching shader was slow, and required optimizations
- **Challenges:** Getting segmentation data from TotalSegmentator through 3D Slicer into Unity with correct coordinate alignment required multiple conversion steps, each introducing potential errors
- **Fork math:** 3D angle comparison doesn't work for a 1D twist control. Had to project into the perpendicular plane at each fork, before the math was solvable
- **Coordinate system pain:** DICOM, Unity, and VMTK all use different axis conventions. Spent significant time diagnosing which axis was up vs cerebral in scene space
- Every one of these challenges forced a deeper understanding of the actual problem. The final system is more accurate, more defensible, and more clinically meaningful than what we originally planned

Resources

- **3D Slicer:** <https://www.slicer.org>
- **SlicerVMTK:** <https://github.com/vmtk/SlicerExtension-VMTK>
- **TotalSegmentator:** <https://github.com/wasserth/TotalSegmentator>
- **paper:** <https://doi.org/10.1148/ryai.230024>
- **VMTK:** <http://www.vmtk.org>
- **DICOM data**

Team & Contributions

- **Alex**

- Custom HLSL raymarching shader with AABB acceleration for real-time Quest performance
- DICOM volume loader and segmentation pipeline
- 6 render modes, threshold controls, interactive UI panel

- **Ezra**

- VMTK centerline graph and new system structure
- Fork puzzle navigation with 2D perpendicular plane math
- Two-phase catheter simulation, state machine, fluoroscopy controller, clot placement
- Haptic system, simulation UI

Thank you!

Any questions?
